

Name: Class: Date:

Lesson 2: Change in populations

Worksheet:

Wildlife smarties

Before we can measure the human impact on a species, we need to determine the initial population. We can demonstrate this by marking and recapturing wild smarties.

Materials

- large bag of Smarties® (or similar shaped, sugar-coated chocolate confectionary)
- small bag of M&M's® (or similar, but must be a variant on Smarties)
- bowl
- spoon

Method

1. Place all the Smarties in the bowl. This is your ecosystem.
2. Capture a small handful of the 'wild' smarties with the spoon and count.
3. Write this number down. (Do not return the smarties to the bowl.)
4. Count out the same number of M&M's and place these in the bowl ecosystem. (These are the animals that have been captured, tagged and released.)
5. Mix the wild smarties with the M&M's thoroughly with the spoon.
6. Use the spoon to take another sample of the smartie and M&M's mix.
7. Count the number of smarties and the number of M&M's in the sample.
8. Write down the total number of lollies there were in this sample. This is the sample size.

Questions

1. How many M&M's were in the final sample? _____
2. Use the formula below to determine how many wild smarties there are in the bowl ecosystem.

$$\text{Number of smarties} = \frac{\text{total number of M\&M's added initially} \times \text{sample size}}{\text{number of M\&M's in the final sample}}$$

$$\text{Number of smarties} = \underline{\hspace{2cm}}$$

This method of marking and recapturing does have some difficulties.

3. What would happen if the marked animals migrated out of that ecosystem?

4. How would the estimated number of smarties be affected if the tagging slowed the animals down so that they were easier to capture (by you or predators)?